# Curriculum Vitae – Rohan Paleja, Ph.D.


## Personal Information

Rohan Paleja
Technical Staff Researcher at MIT Lincoln Laboratory
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## Professional Objective

I focus on developing new computational methods to support robot learning and human-robot collaboration in the diverse and unstructured environments that will be encountered these agents in the real world. Ultimately, I would like to develop novel technologies to helps robots, whether they be household assistive robots or autonomous vehicles, enter into our daily lives.

## Education

**Ph.D. Georgia Institute of Technology, Atlanta GA.** in Robotics. 2018 – 2023
Thesis Title : *Interpretable Artificial Intelligence for Personalized Human-Robot Collaboration.*
Committee : Dr. Matthew Gombolay (Chair), Dr. Peter Stone, Dr. Seth Hutchinson, Dr. Dorsa Sadigh, and Dr. Harish Ravichandar. More details at `www.rohanpaleja.com`.

**M.Sc. Rutgers University, New Brunswick NJ.** in Mechanical Engineering. 2017 – 2018.
Thesis title : *Viability and Performance of Indoor Mapping Using the Velodyne VLP-16 LiDAR.*
Committee : Dr. F. Javier Diez-Garias (Chair), Dr. Haim Baruh, Dr. Aaron Mazzeo.

**B.Sc. Rutgers University, New Brunswick NJ.** in Mechanical Engineering. 2014 – 2017.
*Magna Cum Laude.*

## Awards

**€1000 Sponsorship Award for our ICRA 2023 Explainable Robotics Workshop**, Artificial Intelligence Journal, 2023.
**Interactive Computing Graduate Teaching Assistant of the Year**, Georgia Institute of Technology, 2021.
**Best Workshop Paper Award**, International Conference on Computer Vision (ICCV) Workshop on Multi-Agent Interaction and Relational Reasoning (MAIR2), 2021.
**Best Paper Finalist Award**, Conference of Robot Learning (CoRL), 2020.
**Technology Ventures Award**, Rutgers University, 2016.
**James J. Slade Research Scholar Award**, Rutgers University, 2016.
**General Engineering Scholarship**, Rutgers University, 2015.

## Research Experience

**Technical Staff Researcher. Artificial Intelligence Technology Group, MIT Lincoln Laboratory** Aug 2023–Present.

- Investigating new approaches to help operators understand, program, utilize, and safely team with autonomous systems, ranging from large language models to symbolic agents.
- Leading research teams to study new topics in human-machine teaming.
- Interfaced with sponsors and researchers across the Department of Defense (DoD), presenting research updates and promoting collaboration.

**Graduate Research Assistant, Cognitive Optimization and Relational (CORE) Robotics Lab, Georgia Tech** Aug 2018 – Aug 2023.

- Published more than fifteen publications across the fields of Explainable AI, Interactive Robot Learning, and Multi-Agent Coordination to prestigious conferences and top-tier journal venues, including NeurIPS, AAAI, AAMAS, HRI, RSS, CoRL, IROS, and RA-L.
- Participated in multiple research efforts simultaneously, helping lead novel research directions, facilitate robot demos, conduct human-subject studies, and implement new algorithms and baselines.
- Mentored more than fifteen graduate and undergraduate students across various projects, with over ten contributing to a publication at a top-tier conference or journal.
- Facilitated monthly/quarterly sponsor interactions and technical discussions with Ford Motor Company, MIT Lincoln Laboratory, the Office of Naval Research, and Sandia National Laboratory.

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| **Summer Research Intern. Advanced Concepts and Technologies Group, MIT Lincoln Laboratory** | May 2022 – Aug 2022. |
| **Summer Research Intern. Advanced Concepts and Technologies Group, MIT Lincoln Laboratory** | Jun 2019 – Aug 2019. |
| **Research Assistant, Applied Fluids Laboratory, Rutgers University** | Jan 2016 – Aug 2018 |

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## Publications

### Conference Proceedings -

- Ye*, S., Natarajan*, M., Wu*, Z., Paleja, R., Chen, L., and Gombolay, M. (2023). "Learning Models of Adversarial Agent Behavior under Partial Observability" In Proceedings of the International Conference on Intelligent Robots and Systems (IROS) [**43.3.% Acceptance Rate**]
- Lee*, K., Krishna*, A., Zaidi, Z., Paleja, R., Chen, L., Hedlund-Botti, E., Schrum, M., and Gombolay, M. (2023) "The Effect of Robot Skill Level and Communication in Rapid, Proximate Human-Robot Collaboration" In Proceedings of the Conference of Human-Robot Interaction (HRI). [**25.2% Acceptance Rate**]
- Chen*, L., Jayanthi*, S., Paleja, R., Martin, D., Zakharov, V., and Gombolay, M. (2022) "Fast Lifelong Adaptive Inverse Reinforcement Learning from Crowdsourced Demonstrations" In Proceedings of Conference on Robot Learning (CoRL). [**39% Acceptance Rate**]
- Paleja*, R., Niu*, Y., Silva, A., Ritchie, C., Choi, S., and Gombolay, M. (2022) "Learning Interpretable, High-Performing Policies for Autonomous Driving" In Proceedings of the Robotics : Science and Systems Conference (RSS). [**32% Acceptance Rate**]
- Seraj*, E., Wang*, Paleja*, R., Z., Martin, D., Sklar, M., Patel, A., and Gombolay, M. (2022) "Learning Efficient Diverse Communication for Cooperative Heterogeneous Teaming" In Proceedings of the Conference on Autonomous Agents and Multiagent Systems (AAMAS). [**26% Acceptance Rate**]
- Paleja, R., Ghuy, M., Ranawaka, N., and Gombolay, M. (2021) "The Utility of Explainable AI in Ad Hoc Human-Machine Teaming" In Proceedings of the Conference on Neural Information Processing Systems (NeurIPS). [**26% Acceptance Rate**]
- Schrum, M., Neville, G., Johnson, M., Moorman, N., Paleja, R., Feigh, K., and Gombolay, M. (2021) "Effects of Social Factors and Team Dynamics on Adoption of Collaborative Robot Autonomy." In Proceedings of the Conference of Human-Robot Interaction (HRI). [**23% Acceptance Rate**]
- Paleja, R., Silva, A., Chen, L., and Gombolay, M. (2020) "Interpretable and Personalized Apprenticeship Scheduling : Learning Interpretable Scheduling Policies from Heterogeneous User Demonstrations." In Proceedings of the Conference on Neural Information Processing Systems (NeurIPS). [**20% Acceptance Rate**]
- Paleja*, R., Niu*, Y., and Gombolay, M. (2021) "Multi-Agent Reinforcement Learning with Graph-Attention Communication." In Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS). [**25% Acceptance Rate]**
- Chen, L., Paleja, R., Ghuy, L., and Gombolay, M. (2020) "Joint Goal and Strategy Inference across Heterogeneous Demonstrators via Reward Network Distillation." In Proceedings of the Conference of Human-Robot Interaction (HRI). [**24% Acceptance Rate**]
- Chen, L., Paleja, R., and Gombolay, M.. (2020) "Learning from Suboptimal Demonstration via Self-Supervised Reward Regression." In Proceedings of the Conference on Robot Learning (CoRL).[**Best Paper Finalist**] [**Plenary Talk**][**34% Acceptance Rate**]

### Journal Papers -

- Paleja*, R., Chen*, L., Niu*, Y., Silva, A., Li, Z., Zhang, S., Ritchie, C., Choi, S., Chang, K.C., Tseng, H.E., Wang, Y., Nageshrao, S., and Gombolay, M. (2023). "Learning Interpretable, High-Performance Policies for Continuous Control." **[In Review]**
- Seraj*, E., Paleja*, R., Pimentel, L., Lee, K.M., Martin, D., Sklar, M., Zhang, J., Kakish, Z., and Gombolay, M. (2023). "Heterogeneous Policy Networks for Composite Robot Team Communication and Coordination." *IEEE Transactions on Robotics (T-Ro)*, Volume 40, pages 3833-3849.
- Natarajan*, M., Seraj*, E., Altundas*, B., Paleja*, R., Ye*, S., Chen*, L., Jensen, R., Chang, K.C., and Gombolay, M. (2023). "Human-Robot Teaming : Grand Challenges." *Current Robotics Reports*, pp.1-20.
- Zaidi, Z., Martin, D., Belles, N., Zakharov, V., Krishna, A., Lee, K.M., Wagstaff, P., Naik, S., Sklar, M., Choi, S., Kakehi, Y., Patil, R., Mallemadugula, D., Pesce, F., Wilson, P., Hom, W., Diamond, M., Zhao, B., Moorman, N., Paleja, R., Chen, L., Seraj, E., and Gombolay, M. (2022). "Athletic Mobile Manipulator System for Robotic Wheelchair Tennis." *IEEE Robotics and Automation Letters (RA-L)*, Volume 8, Issue 4, pages 2245-2252.

— Dias, D., Zenati, M., Srey, R., Arney, D., Chen, L., Paleja, R., Kennedy-Metz, L., and Gombolay, M.. (2021) "Using Machine Learning to Predict Perfusionists' Critical Decision-Making during Cardiac Surgery." *Computer Methods in Biomechanics and Biomedical Engineering. Imaging & Visualization*, 10(3), 308-312.

**Workshop Papers, Symposium Papers, and Doctoral Consortia** -

— Sreeramdass, V., Paleja, R. R., Chen, L., van Waveren, S., and Gombolay, M. (2023). "Generalized Behavior Learning from Diverse Demonstrations." In Proceedings of the CoRL Workshop on Out-of-Distribution Generalization in Robotics.

— Wu, Z., Ye, S., Natarajan, M., Chen, L., Paleja, R., and Gombolay, M. (2023). "Adversarial Search and Tracking with Multiagent Reinforcement Learning in Sparsely Observable Environments" In Proceedings of the International Symposium on Multi-Robot and Multi-Agent Systems (MRS) 2023 [**35% Acceptance Rate**]

— Pimentel, L.*, Paleja, R.*, Wang, Z., Seraj, E., Pagan, J., and Gombolay, M. (2022). "Scaling Multi-Agent Reinforcement Learning via State Upsampling." In Proceedings of the Robotics Science and Systems Workshop on Scaling Robot Learning (RSS22-SRL).

— Paleja, R., and Gombolay, M. (2022). "Mutual Understanding in Human-Machine Teaming." In Proceedings of the Association for the Advancement of Artificial Intelligence Conference (AAAI) Doctoral Consortium.

— Niu*, Y., Paleja*, R., and Gombolay, M. (2021) "Multi-Agent Graph-Attention Communication and Teaming." In Proceedings of the ICCV 2021 Workshop on Multi-Agent Interaction and Relational Reasoning. **[Spotlight Talk] [Best Workshop Paper Award]**

— Chen, L., Paleja, R., and Gombolay, M.. (2021) "Towards Sample-efficient Apprenticeship Learning from Suboptimal Demonstration." In Proceedings of Artificial Intelligence for Human-Robot Interaction (AI-HRI), AAAI Fall Symposium Series.

— Paleja, R., Silva, A., Chen, L., and Gombolay, M. (2021) "Interpretable and Personalized Apprenticeship Scheduling : Learning Interpretable Scheduling Policies from Heterogeneous User Demonstrations." In Proceedings of the AAMAS Autonomous Robots and Multirobot Systems (ARMS) Workshop.

— Paleja, R., and Gombolay, M. (2020) "Heterogeneous Learning from Demonstration." In Proceedings of the Conference of Human-Robot Interaction (HRI) Pioneers Workshop. [**32% Acceptance Rate**]

**Thesis** -

— Paleja, R. (2023) "Interpretable Artificial Intelligence for Personalized Human-Robot Collaboration." *Ph.D. Thesis*, Georgia Institute of Technology.

— Paleja, R. (2018) "Viability and Performance of Indoor Mapping using the Velodyne VLP-16 LiDAR." *M.Sc. Thesis*, Rutgers University.

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## Teaching Experience

**Teaching Assistantship**

— Introduction to Robotics Research (CS 7785), School of Interactive Computing (IC), Georgia Institute of Technology (Spring 2022) | Supervisor : Dr. Sean Wilson

— Robot Intelligence : Planning (CS 7469-A) – Graduate Section, School of Interactive Computing (IC), Georgia Institute of Technology (Fall 2020) | Supervisor : Prof. Matthew C. Gombolay

— Robot Intelligence : Planning (CS 4649-A) – Undergraduate Section, School of Interactive Computing (IC), Georgia Institute of Technology (Fall 2020) | Supervisor : Prof. Matthew C. Gombolay

— Dynamics of Rigid Bodies (ME 2202), School of Mechanical Engineering (ME), Georgia Institute of Technology (Summer 2020) | Supervisor : Prof. Nader Sadegh

— Alternative Energy Systems (ME 474), School of Mechanical Engineering (ME), Rutgers University (Fall 2017) | Supervisor : Prof. Sara Moghtadernejad

— Aerospace Propulsion (ME 459), School of Mechanical Engineering (ME), Rutgers University (Spring 2018) | Supervisor : Prof. Doyle Knight

**Advising & Mentorship**

— Mentor for the MIT Beaver Works CRE[AT]E Assistive Technology Challenge

— Research Mentorship (9 Graduate and 7 Undergraduate Students)

  — Varshith Sreeramdas, M.Sc. Student, Georgia Tech. January 2022 – August 2022

  — Arjun Krishna, M.Sc. Student, Georgia Tech. May 2022–May 2023 (*Co-First Author Paper* : [Krishna et al. ; HRI'23])

  — Kin Man Lee, M.Sc. Student, Georgia Tech. May 2022–May 2023 (*Co- First Author Paper :* [Krishna et al. ; HRI'23])

  — Daniel Martin, M.Sc. Student, Georgia Tech. August 2021–May 2022 (*Co-Author on Two Papers :* [Seraj et al. ; AAMAS'22, Zaidi et al. ; ICRA-RAL])

- Matthew Sklar, M.Sc. Student, Georgia Tech. May 2021–December 2021 (*Co-Author on Two Papers :* [Seraj et al. ; AAMAS'22, Zaidi et al. ; ICRA RA-L])
- Luis Pimentel, M.Sc. Student, Georgia Tech. January 2022–August 2023 (*First-Author Workshop Paper :* [Pimentel et al. ; RSS22-SRL])
- Michael Munje, M.Sc. Student, Georgia Tech. September 2022–May 2023
- John Zhang, M.Sc. Student, Georgia Tech. January 2022–August 2022
- Yaru Niu, M.Sc. Student, Georgia Tech. August 2020-May 2022 (*Two Co-First Author Papers :* [Niu et al. ; AAMAS'21, Paleja et al. ; RSS'22])
- Chace Ritchie, Undergraduate Student, University of Kentucky. *Under the NSF SURE Robotics Program.* May 2021–August 2021 (*Co-Author Paper :* [Paleja et al. ; RSS'22])
- Ruturaj Patil, Undergraduate Student, Georgia Tech. May 2021–August 2021 (*Co-Author on Paper :* [Zaidi et al. ; ICRA RA-L] )
- Sugju Choi, Undergraduate Researcher, Georgia Tech. May 2021–August 2021 (*Co-Author on Paper :* [Paleja et al. ; RSS'22])
- Nadun Ranawaka Arachchige, Undergraduate Student, Georgia Tech. January 2021–May 2021 (*Co-Author on Paper :* [Paleja et al. ; NeurIPS'21])
- William Silva, Undergraduate Student, Georgia Tech. May 2021–August 2021
- Erik Scarlatescu, Undergraduate Student, Georgia Tech. August 2022–December 2022
- Lokranjan Laksmikanthan, Undergraduate Student, Georgia Tech. May 2022–August 2022
- Sergey Savelyev, Undergraduate Student, Georgia Tech. January 2019–May 2019 (*Undergraduate Thesis* : Mastering Reconnaissance Blind Chess with Reinforcement Learning)

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## Academic Service

**Leadership**

- **Workshop Organizer**, CoRL 2023 Workshop on Robot Learning in Athletics, Atlanta, Georgia, USA.
- **Public Relations Vice President**, Executive Board of the Robotics Graduate Student Organization, Georgia Institute of Technology.
- **Workshop Organizer**, ICRA 2023 Workshop on Explainable Robotics, London, United Kingdom.
- **Sponsorship Chair**, Human-Robot Interaction (HRI) 2020 Pioneers Workshop, Cambridge, United Kingdom.

**Technical Reviewer**

- Robotics : Science and Systems (RSS)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- International Conference on Human-Robot Interaction (HRI)
- International Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Artificial Intelligence and Statistics (AISTATS)
- Association for the Advancement of Artificial Intelligence Conference (AAAI)
  - Special Track on Safe, Robust and Responsible AI (SRRAI)
  - Special Track on AI for Social Impact (AISI)
- Autonomous Agents and Multiagent Systems (AAMAS)
- International Conference on Robotics and Automation (ICRA)
- Robotics : Science and Systems Conference (RSS)
- International Journal of Human-Computer Interaction (IJHCI)
- International Conference on Advanced Robotics (ICAR)
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- IEEE Transactions on Cybernetics
- International Conference on Robot & Human Interactive Communication (ROMAN)

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## Memberships

**IEEE Student Member**
**RoboGrads, Robotics Graduate Student Organization**
**Pi Tau, Mechanical Engineering Honor Society**
**American Society of Mechanical Engineers**

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## Selected Press Coverage

*This tennis-playing robot could one day win Wimbledon* | USA Today | IEEE Spectrum | Tyler Morning Paragraph | News on the Neck | The News Times | Talker News | Daily Mail | Metro UK | Independent Record | Kenosha News | Tennis

**Georgia Tech College of Computing** (US) *Tennis Robot Could Pave Way for Advancement in Fast-Movement Robotics* Video | Blog | Mashable | IOT World Today | Interesting Engineering | Watson
**Georgia Tech College of Computing** (US) *Georgia Tech Researchers Teach a Robot How to Improve at Ping Pong on Its Own* Video | Blog
**Georgia Tech Daily Digest** (US) *Georgia Tech Researchers Use Table Tennis to Understand Human-Robot Dynamics* Blog

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## Invited Talks

**Safe AI Lab Seminar**, Carnegie Mellon University, *Virtual. (February, 2024)* **Transformative Science and Technology Lecture Series**, Temple University. *Philadelphia, PA, 19112, USA (September 2023)*
**Robot Learning Seminar**, Mila. *Virtual. (June 2023)*
**Robotics Talk**, Brown University. *Providence, RI 02912, USA (February 2023)*

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## References

References can be provided upon request